

# FA6

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**GitHub link:** <https://github.com/aldwinmontilla661-dot/Aldwin-Benidict-Montilla/tree/main/APM1110-FA6>

## I. Geometric Distribution

Provide an R code for the geometric distribution. The geometric distribution is a probability distribution that models the number of trials required to achieve the first success in a sequence of Bernoulli trials, where each trial has a constant probability of success.

1. Set the probability of success:  $p \leftarrow 0.2$

```
p <- 0.2
```

2. Generate 1000 random variables from the geometric distribution.

```
# We will have to add 1 because rgeom() counts failures, but we want the total number of trials.  
set.seed(42) # Optional: ensures identical results every time you run it  
x <- rgeom(1000, prob = p) + 1
```

3. Calculate basic statistics:

```
mean_x <- mean(x)  
var_x <- var(x)  
sd_x <- sd(x)
```

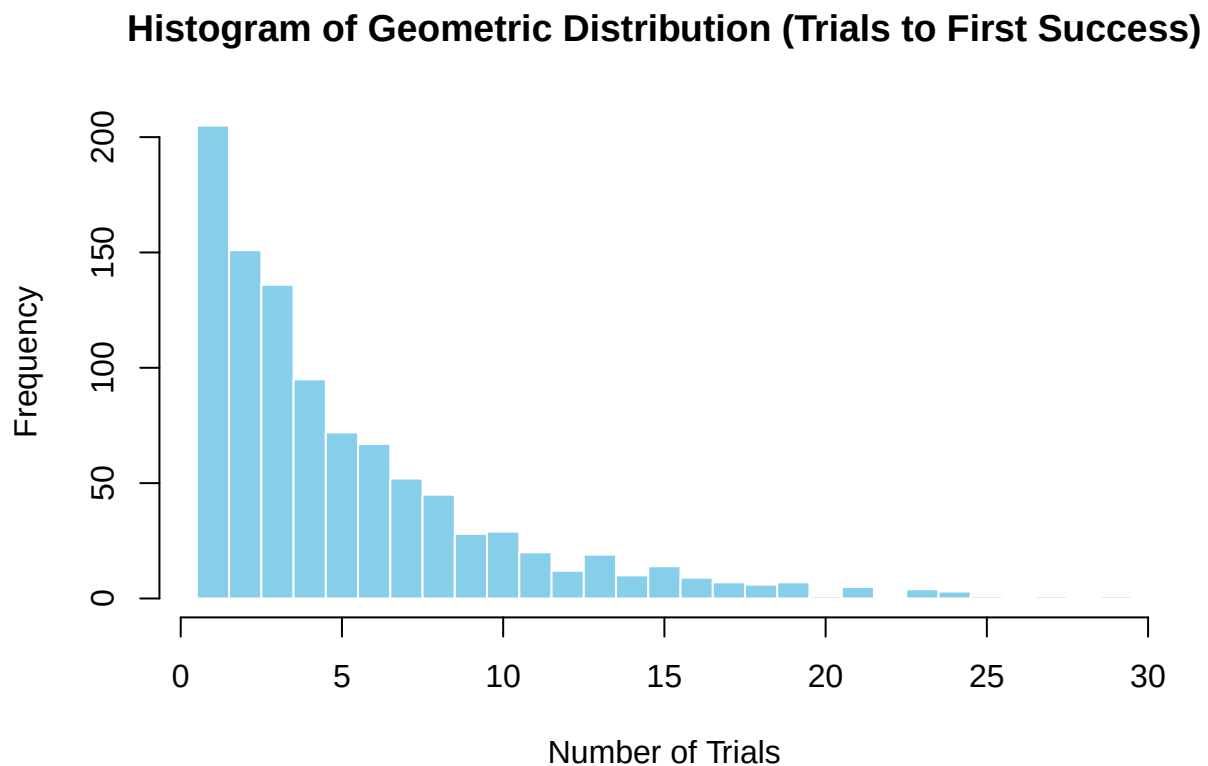
4. Print the results in item 3.

```
cat("Number of trials required to achieve first success:\n",  
    sprintf("%.2f\n", mean_x),  
    sprintf("%.2f\n", var_x),  
    sprintf("%.2f\n", sd_x))
```

```
## Number of trials required to achieve first success:  
## • Mean: 5.16  
## • Variance: 21.69  
## • Standard deviation: 4.66
```

## 5. Plot the histogram of the results.

```
hist(x,  
     breaks = seq(0.5, max(x) + 0.5, by = 1),  
     col = "skyblue",  
     border = "white",  
     main = "Histogram of Geometric Distribution (Trials to First Success)",  
     xlab = "Number of Trials",  
     ylab = "Frequency")
```



## II. Hypergeometric Distribution

Consider a plant manufacturing IC chips of which 10% are expected to be defective. The chips are packed in boxes for export. Before transportation, a sample is drawn from each box. Estimate the probability that the sample contains more than 10% defectives, when:

1. A sample of 10 is selected from a box of 40;

```
# Population size.
N = 40

# Defective chips in the population.
K = 0.1 * N #10% of 40 (population).

# Non-defective chips in population.
non_def = N - K #population minus the defective chips.

# Sample size.
n = 10

# We want  $P(X > 1)$ , which is  $1 - P(X \leq 1)$ .
# We will use the built-in hypergeometric distribution function (phyper) to calculate the exact probability.
# R's phyper(x, K, N-K, n) calculates  $P(X \leq x)$ .

prob_box_40 <- 1 - phyper(1, m = K, n = non_def, k = 10)
```

Print the results

```
cat("Hypergeometric Distribution Probabilities (Sample > 10% Defective):\n",
    sprintf("1. Box of 40 chips:   %.4f (or %.2f%%)\n", prob_box_40, prob_box_40 * 100))

## Hypergeometric Distribution Probabilities (Sample > 10% Defective):
## 1. Box of 40 chips:   0.2559 (or 25.59%)
```

2. A sample of 10 is selected from a box of 5000.

```
# Population size.
N = 5000

# Defective chips in the population.
K = 0.1 * N #10% of 5000 (population).

# Non-defective chips in population.
non_def = N - K #population minus the defective chips.

# Sample size.
n = 10
```

```
# We will use the same method of calculation for the first scenario.
prob_box_5000 <- 1 - phyper(1, m = K, n = non_def, k = 10)
```

Print the results

```
cat("Hypergeometric Distribution Probabilities (Sample > 10% Defective):\n",
    sprintf("2. Box of 5000 chips: %.4f (or %.2f%%)\n", prob_box_5000, prob_box_5000 * 100))
```

```
## Hypergeometric Distribution Probabilities (Sample > 10% Defective):
## 2. Box of 5000 chips: 0.2639 (or 26.39%)
```

## Interpretation

In the box of 40, every time you pull out a chip, you significantly alter the remaining pool. If you pull a clean chip, the concentration of defective chips left behind goes up. This dependency “dampens” extreme variance, making the probability of pulling multiple defectives slightly lower (25.59%).

In the box of 5000, the pool is so massive that drawing 10 chips barely modifies the remaining ratio. The trials act almost completely independent of one another. As a result, the probability rises slightly to 26.39%, which perfectly converges toward a standard Binomial Distribution calculation ( $B(10, 0.10)$ ).